Ariel Szekely

Contact Information

Phone	+1 (817) 876-2431
Email	arielszekely@gmail.com
Homepage	www.arielck.com
GitHub	github.com/ArielSzekely

Education

2020-present **Ph.D. in Computer Science**, *Massachusetts Institute of Technology*, Cambridge, MA

2020-2022 **M.S. in Computer Science**, *Massachusetts Institute of Technology*, Cambridge, MA,
Thesis: *σOS: Elastic Realms for Multi-Tenant Cloud Computing*
GPA: 5.0/5.0

2016-2020 **B.S. in Computer Science (Honors)**, *The University of Texas at Austin*, Austin, TX,
Thesis: *Exploring GPU Timing Side-Channels*
GPA: 3.94/4.0

Interests

Distributed Systems, Operating Systems, Security, Concurrency.

Professional Experience

2023 **Student Researcher**, *Google Systems Research Group*, Seattle, WA
Designed and evaluated a system for TPU disaggregation for Ads model serving.

2019 **Research Assistant**, *The University of Texas at Austin*, Austin, TX
Developed the Telekine GPU timing side-channel attack for our NSDI '20 paper.

2018 **Software Engineer Intern**, *Bloomberg L.P.*, New York City, NY
Back-end development.

2017 **Software Engineer Intern**, *AT&T*, Dallas, TX
Full-stack development.

Publications

Eurosys '27 (under submission) **Fast end-to-end cloud application cold-start with co-sandboxes**, **Ariel Szekely**, *Hannah Gross, Robert Morris, Frans Kaashoek*
Designed and implemented co-sandboxes, a new scriptable interface for cloud developers to specify their application's initialization routine to the provider in order to overlap application initialization with infrastructure setup costs. Co-sandboxes reduce cold-start latency of serverless applications by 1.6-1.8X and microservices like etcd and memcached by 1.7X.

SOSP '24 **Unifying serverless and microservice workloads with SigmaOS**, **Ariel Szekely**, *Adam Belay, Robert Morris, M. Frans Kaashoek*
Designed and implemented SigmaOS, a novel cloud operating system and cluster manager which supports both microservices and serverless applications. SigmaOS's API enables applications to warm-start in under 2ms and cold-start in under 8ms with strong isolation in a lightweight container.
[GitHub]

ASPLOS '23 **Towards a machine learning-assisted kernel with LAKE**, *Henrique Fingler, Isha Tarte, Hangchen Yu,* **Ariel Szekely**, *Bodun Hu, Aditya Akella, Christopher J. Rossbach*
Rewrote the Linux Kernel Samepage Merging (KSM) module to benefit from hardware acceleration.

NSDI '20 **Telekine: Secure computing with cloud GPUs**, *Tyler Hunt, Zhipeng Jia, Vance Miller,* **Ariel Szekely**, *Yige Hu, Christopher J. Rossbach, Emmett Witchel*
Built a timing side-channel attack against a GPU Trusted Execution Environment (TEE), allowing the attacker to learn the class of images passed through ResNet by timing the GPU kernels.

Awards

2022 NSF Graduate Research Fellowship

2020 MIT Lemelson Presidential Fellowship

2020 MIT EECS Departmental Fellowship

2020 Dean's Honored Graduate, UT Austin College of Natural Sciences

2020 Research Distinction, UT Austin College of Natural Sciences

2020 NSF Graduate Research Fellowship Honorable Mention

2020 CRA Outstanding Undergraduate Researcher Honorable Mention

2019 NSF REU

2019 UT Endowed Presidential Scholarship

2018,19,20 UT College of Natural Sciences College Scholar

2018 Louis E. Rosier Memorial Scholarship

2016 UT CNS Freshman Scholarship

2016 National Merit Scholarship

Mentorship and Advising

2026 **Nicolas Camenisch**, *MIT*, M.Eng. Thesis
Thesis title: *Fork Around and Find Out: Accelerating Dynamic Language Runtimes in SigmaOS via Lightweight Container Forking*

2026 **Nathan Mustafa**, *MIT*, M.Eng. Thesis
Thesis title: *Efficient RDMA for SigmaOS*

2025 **Ivy Wu**, *MIT*, M.Eng. Thesis
Thesis title: *Interposing the syscall boundary: Transparent python execution in SigmaOS*

2025 **Ryan Chang**, *MIT*, M.Eng. Thesis
Thesis title: *Optimizing SigmaOS for efficient orchestration of fault-tolerant, burst-parallel workloads*

2025 **Freddie Tang**, *MIT*, M.Eng. Thesis
Thesis title: *Accelerating Burst Parallelism of SigmaOS processes with CRIU*

2025 **Katie Liu**, *MIT*, M.Eng. Thesis
Thesis title: *Enabling efficient ML inference in SigmaOS with model-aware scheduling*

2023 **Yizheng He**, *MIT*, M.Sc. Thesis
Thesis title: *Evaluating sigmaos with kubernetes for orchestrating microservice and serverless applications*

Service

2021 **SOSP '21 Artifact Evaluation Committee**, *SOSP 2021*, Kolbenz, Germany

2022-present **EECS Communication Lab Fellow**, *Massachusetts Institute of Technology*, Cambridge, MA

2020-2022 **MIT Graduate Application Assistance Program Mentor**, *Massachusetts Institute of Technology*, Cambridge, MA

Teaching Experience

2023 **TA: Computer Systems Security**, *Massachusetts Institute of Technology*
Instructors: Nickolai Zeldovich

2022 **TA: Operating Systems Engineering**, *Massachusetts Institute of Technology*
Instructors: Frans Kaashoek, Robert Morris

2020 **TA: Concurrency (Honors)**, *The University of Texas at Austin*
Instructor: Christopher J. Rossbach

2020 **TA: Intro to CS Research (Honors)**, *The University of Texas at Austin*
Instructor: Calvin Lin

2019 **TA: Intro to CS Research (Honors)**, *The University of Texas at Austin*
Instructor: Calvin Lin

Languages

English Native
Spanish Native
French Fluent